HARIKRISHNA MEKALA

Senior Software Engineer · Trading Backends · Derivatives Markets · AI/ML Platforms

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8 years · **$1T+ TPV** settlement scale · **23 global exchanges** · **1M+ AI calls** · **Agentic AI** · **4.5M+ users**

SUMMARY

Senior Software Engineer with **8 years** building **low-latency, fault-tolerant trading and payments backends** across institutional fintech and AI/ML platforms. Architected a **Trading Backend** routing across **23 global exchanges** over the **FIX Protocol**, designed a **Derivatives Market** layer for custom OTC instruments, drove PayPal's **Payments cloud migration** at **$1T+ TPV scale**, and owned the **OpenAI LLM Gateway** within **PayPal Cosmos** powering **Agentic AI** at **1M+ AI calls**.

EXPERIENCE

KeenAi Global – *Senior Associate*

MAY 2025 - PRESENT **HYBRID, BANGALORE**

- Architected and shipped a **Trading Backend** routing flows across **23 global exchanges** (US, Europe, Singapore) and broker-dealers (**Pershing / NetX360**) over the **FIX Protocol** for institutional equity, options, and OTC structured products.
- Built an end-to-end **real-time order lifecycle** pipeline — placement, ack, partial-fill, execution report, allocation, settlement — with idempotent state machines and exactly-once event semantics on **AWS SQS**.
- Designed and built the **Derivatives Market** data layer on MongoDB — the source-of-truth for **custom OTC instruments** (FCNs, structured notes) — exposing listings, quote-currency, LCM, issuer credit rating, and risk APIs.
- Engineered a **fault-tolerant Pershing webhooks** ingestion pipeline (API Gateway + ACM + SQS FIFO + EventBridge) with at-least-once delivery, DLQ handling, and replay — provisioned end-to-end with **Terraform**.
- Owned the **FIX execution-report consumer** for OTC FCN trades, reconciling **back-to-back NetX360** ↔ **Pershing** flows and matching trade state across upstream brokers and downstream books.
- Executed a **full-stack platform rewrite** from **Spring Boot + MySQL** to **NestJS + MongoDB** with zero downtime, unifying the runtime and dramatically improving developer velocity and query performance.
- Authored **Temporal workflows** for long-running structured-product lifecycle events (issuance, coupon payments, knock-in / knock-out, early redemption) with deterministic retries, durable state, and human-in-the-loop escalation.
- Built a low-latency in-memory **trie + scored-cache** search engine over the global instrument universe, returning **sub-100 ms** ranked suggestions across **millions of instruments**.
- Replaced legacy hardcoded keys with a **Cognito-backed service-to-service auth** layer, centralizing credential issuance, rotation, and audit across internal microservices.
- Built **Vault**, a secure Dropbox-like internal file-storage system with **fine-grained RBAC**, per-Family-Office tenancy, audit logs, and signed-URL delivery for institutional clients.

PayPal – *Software Engineer 3*

SEPTEMBER 2024 - MAY 2025 **HYBRID, BANGALORE**

- Owned the **OpenAI LLM Gateway** within the **PayPal Cosmos** AI/ML Platform, scaling **multi-tenant inference, training, and Agentic AI** infrastructure to **1M+ AI calls** across PayPal's product surfaces.
- Built a unified **LLM Platform** integrating **Azure OpenAI**, enabling production-grade processing of audio, image, and text data for fraud detection, risk scoring, and customer support.
- Hardened the **Data Loss Prevention (DLP)** system, strengthening data security, risk management, and regulatory compliance across PayPal's enterprise data surfaces.
- Implemented **multi-model context management and hot-switching** across **Gemini, ChatGPT, and Claude** with request-time routing for cost / latency / quality tradeoffs.

PayPal – *Software Engineer 2* JUNE 2023 - AUGUST 2024 HYBRID, CHENNAI

- Led the reengineering and **AWS cloud migration** of PayPal's **Payments settlement infrastructure** (**$1T+ annual TPV**) from on-prem to a fault-tolerant distributed system on **DynamoDB + EKS**.
- Built a **real-time snapshotting** solution to accelerate bank settlements, reducing end-of-day processing time and improving daily transaction throughput.
- Spearheaded the migration of a legacy **C++ codebase to Java Spring Boot**, modernizing a critical settlement component and improving maintainability.
- Served as **Scrum Master** for PayPal's **Payments Cloud Migration Team**, driving Agile delivery and continuous improvement throughout the on-prem-to-AWS reengineering.

smallcase – *Software Development Engineer 1* APRIL 2022 - MAY 2023 REMOTE, BANGALORE

- Designed a highly available, **low-latency Stock Scorecard REST API** serving **4.5M+ active users**, powering Tickertape's flagship equity-research surface.
- Implemented proprietary algorithms for **PE ratios, portfolio volatility, and equity-fund ranking** by market cap and allocation — feeding Tickertape's screener and recommendation engine.
- Integrated **24 new market indices** for real-time sector tracking and added regular Mutual Fund coverage, expanding the platform's tradable universe by **50%**.
- Built internal **gRPC** services with **LRU-cache**-backed hot paths to scale Scorecard fan-out to **millions of requests/day**.
- Led the migration of Tickertape's core API to a **pod-level segregated NestJS architecture** for fault isolation and stability.

Aardwolf Security – *Software Engineer* NOVEMBER 2019 - MARCH 2022 REMOTE, LONDON

- Built a highly available, **fault-tolerant content ranking system** with a customer-facing REST API and a high-throughput RSS parsing engine.
- Established the full **CI/CD pipeline** (Jenkins + SonarQube) and implemented automated **firewall-based traffic throttling** to mitigate DDoS attacks.
- Built a **PayPal Payouts**-backed affiliate system and designed a **manager-worker queuing system** for large-scale asynchronous content analysis.

Kloudgin – *Software Engineer* JULY 2019 - NOVEMBER 2019 OFFICE, HYDERABAD

- Architected a modular **Netsuite integration** for the FSM platform and engineered a **distributed multi-threaded data-migration** system for enterprise customers.

Infovity – *Technical Consultant* MAY 2018 - JULY 2019 OFFICE, HYDERABAD

- Contracted to Kloudgin: added video-streaming for crew training, built a **Java iText** wrapper for custom PDF invoices, and shipped a **Docker-based microservices** app for billables / biometrics tracking.

SKILLS

DOMAIN **Trading Systems** · **Derivatives Markets** · **Order Management Systems (OMS)** · **Instrument Master (IMS)** · **FIX Protocol** · **OTC Structured Products** · **Fixed Coupon Notes (FCN)** · Trade Lifecycle · Settlement · Wealth Management · Family Offices · Pershing / NetX360

ARCHITECTURE Distributed Systems · Microservices · Event-Driven Architecture · Fault Tolerance · High Availability · Low-Latency Design · Idempotency · Exactly-Once Semantics · System Design · Performance Tuning

LANGUAGES Java (Spring Boot) · TypeScript / Node.js (NestJS) · Python · Go · C++ (legacy) · Shell

DATA & MESSAGING MongoDB · DynamoDB · MySQL · Redis · Kafka · **AWS SQS** · **EventBridge** · gRPC · REST · **Temporal Workflows**

CLOUD & DEVOPS AWS (EKS, EC2, Lambda, API Gateway, ACM, CodeDeploy, CloudWatch, IAM, **Cognito**) · **Terraform** · Docker · Kubernetes · Jenkins · CI/CD · SonarQube · Sentry

AI / ML **OpenAI LLM Gateway** · LLM Inference & Training · Azure OpenAI · Gemini · ChatGPT · Claude · Multi-Model Context Management

METHODOLOGY Agile / Scrum (Scrum Master) · DevOps · Code Review · Mentoring

INTERESTS Building custom PC workstations · Contributing to open-source projects

EDUCATION

Vishnu Institute of Technology – *Computer Science and Engineering* 2018 BHIMAVARAM, INDIA

- **Cumulative Percentage: 72%** *(Distinction)*.
- Awarded **1st place** in the VEDIC Inter-Collegiate Coding Competition.
- Built a **containerized code compiler** for secure remote code execution, the **real-time ranking system** for the annual college coding competition, and a **Moodle plugin** for automatic lab-record generation.